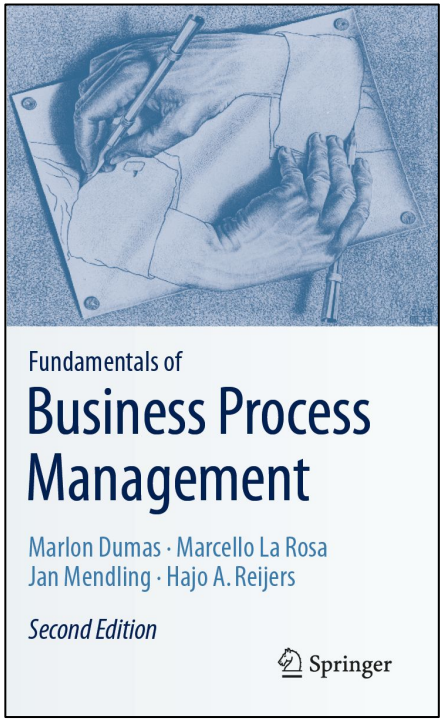
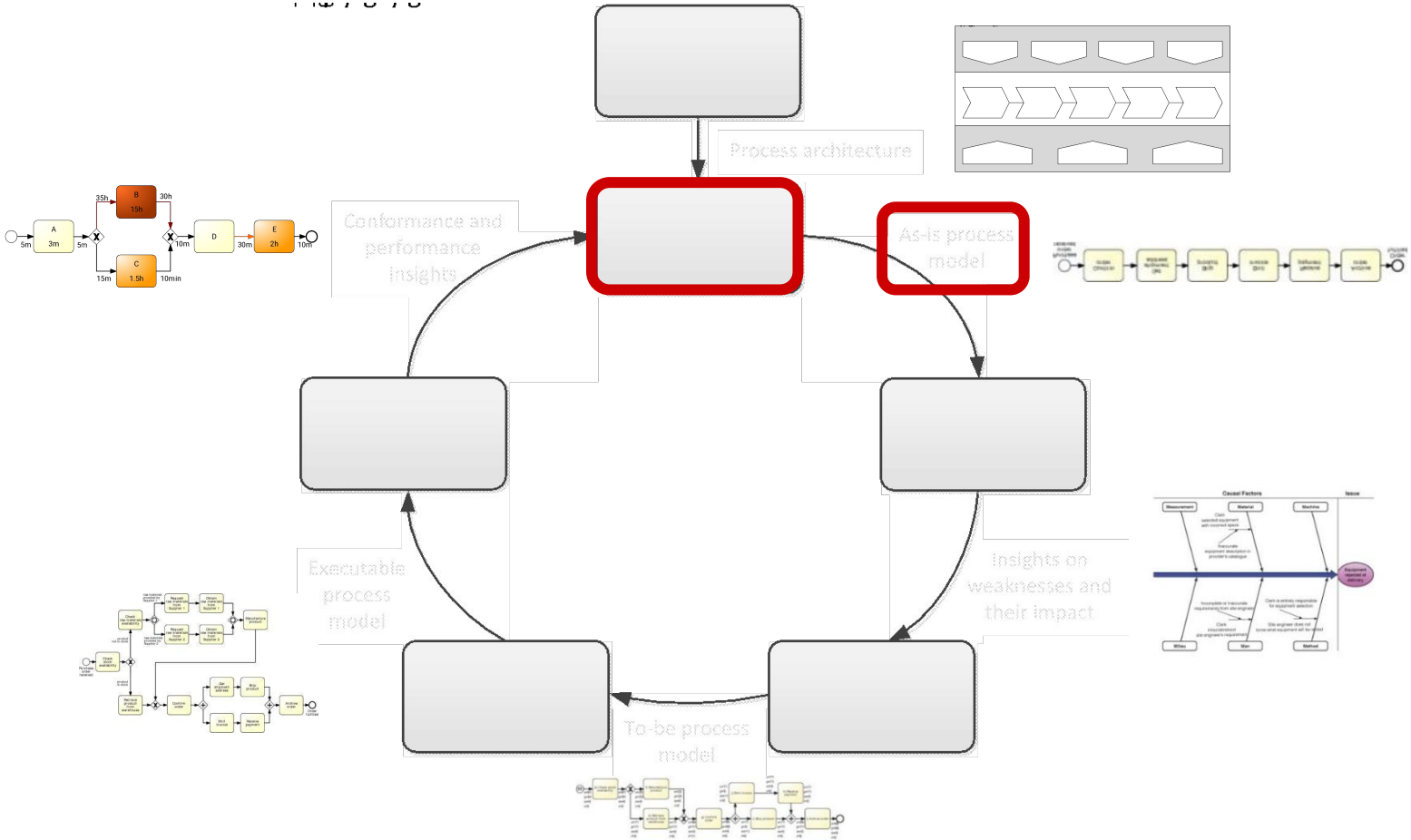
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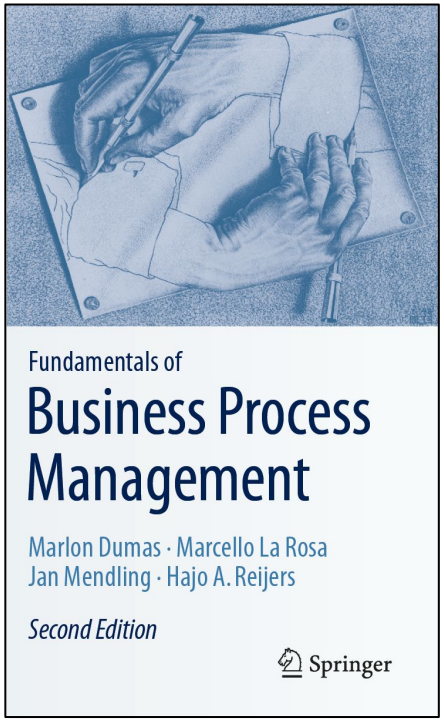
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Process Modeling in the BPM Lifecycle



Chapter 5: Process Discovery



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Process discovery

Four main tasks:

- **Define the setting:** assemble a team that will be responsible for managing the process
- **Gather information:** build an understanding of the process. Different discovery methods available
- **Conduct the modeling:** do the actual modeling
- **Assure model quality:** guarantee that the resulting model meets different quality criteria

done together

Who is involved?



Process analyst



Domain expert

Example: modeling perspective

Consider the following two modelling tasks:

- Modeling the process for ordering books through an online bookstore, from the perspective of the customer.
- Modeling the same process from the perspective of the bookstore.

Which of the two tasks above are you familiar with and why?

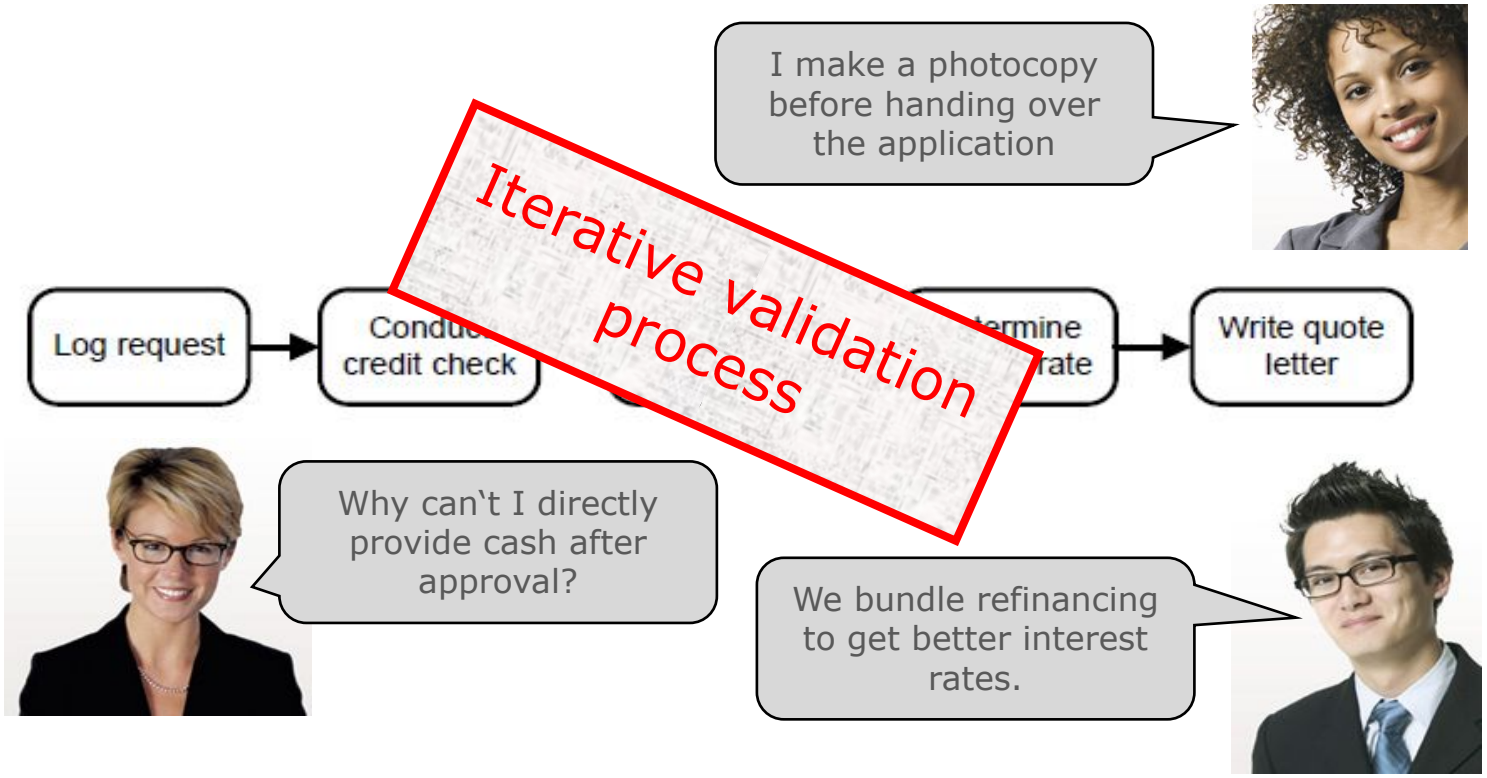


Exercise 5.1

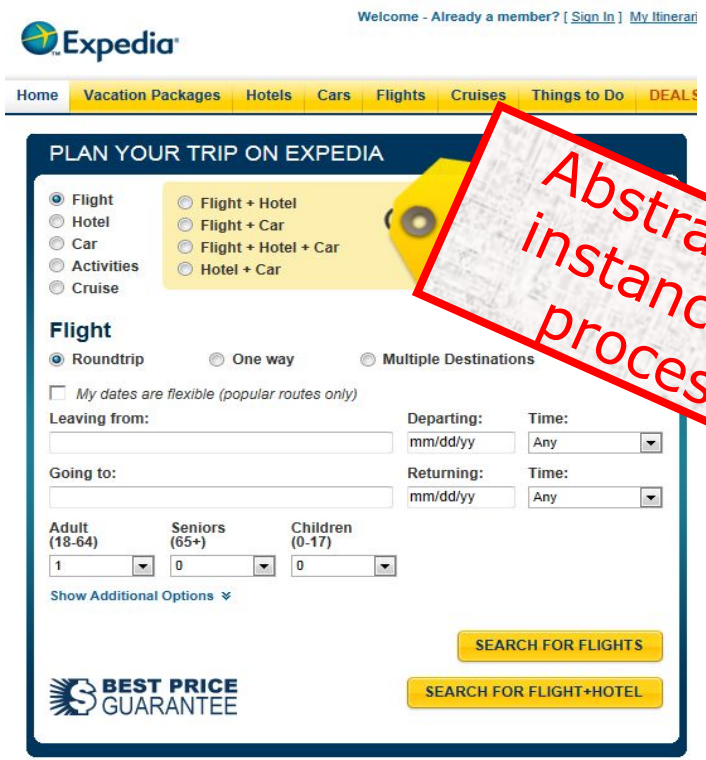
You are the manager of a consulting company and you need to hire a person for the newly signed BPM project with an online bookstore. Consider the following two profiles; who would you hire as a process analyst?

- Mike Miller has ten years of work experience with an online retailer. He has worked in different teams involved with the order-to-cash process of the online retailer.
- Sara Smith has five years of experience working as a process analyst in the banking sector. She is familiar with two different process modeling languages and with several modeling tools.

Challenge 1: Fragmented process knowledge



Challenge 2: Domain experts think on instance level



Abstraction from instance level to process level

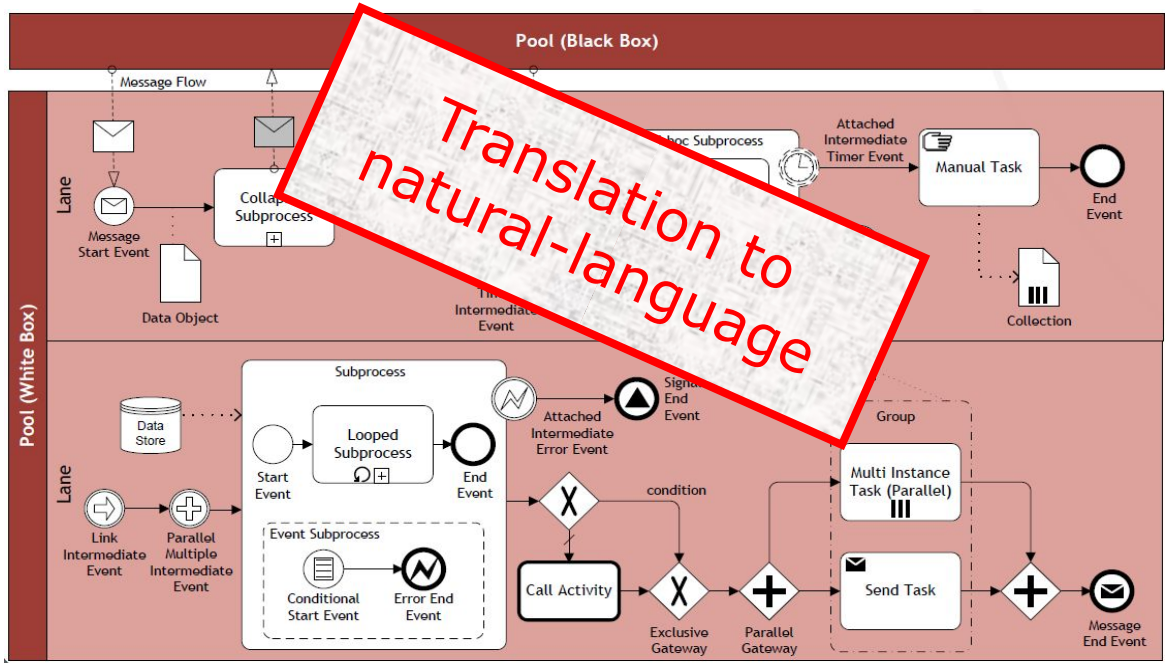
"Every trip is different"

"You cannot really compare. Our customers go to different places in different seasons using different means of transportation"

"We can never do anything exactly in the same way. There are so many special conditions"

Challenge 3: Knowledge of process modeling is uncommon

"Does this diagram correctly show your process?"



What makes a good process analyst?

- Getting the right people on board
- Formulate and test hypotheses
- Identify patterns
- Pay attention to model quality





Exercise 5.2

Consider the order-to-cash process of your preferred online bookstore and assume you have access to three internal resources: a customer relationship manager (who handles sales and reclaims), a warehouse worker (who looks after shipments), and a financial officer (who raises invoices and collects payments). As a process analyst, what questions do you need to ask these domain experts to be able to obtain a complete and systematic view of this process?

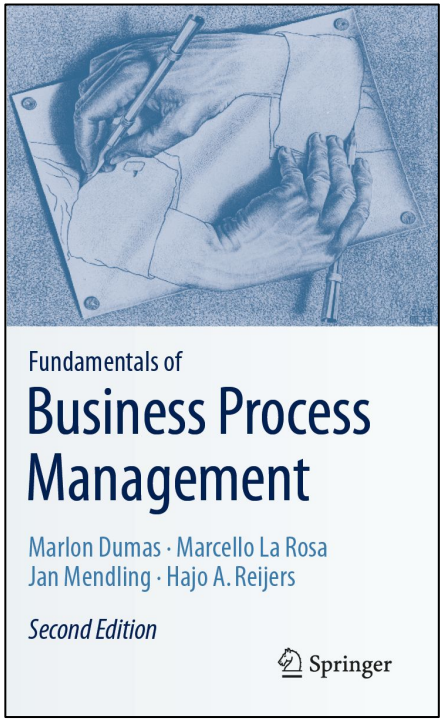
Hint. Think of the different exposure to this process that the three resources have and of the possible conditions, process outcomes, and exceptions that they may have experienced while executing this process.



Profile of an Expert Process Analyst

1. Getting the right people on board
2. Having a set of working hypotheses of how the process is structured at different levels of detail
3. Identifying patterns in the information provided by domain experts
4. Paying attention to model quality

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Process discovery methods

1. **Evidence-based**
   - Document analysis
   - Observation
   - Automated process discovery



2. **Interview-based**



3. **Workshop-based**

Choose one or more on the basis of context and budget

Document Analysis

Documents point to existing roles, activities and business objects:

- Process descriptions (ideal scenario)
- Internal policies
- Organization charts
- Employment plans
- Quality certificate reports
- Glossaries and handbooks
- Forms
- Work instructions…



Could be used to gather information before approaching domain experts.

Potential issues:

- May not be process-oriented and trustworthy
- May require abstraction or refinement

Observation

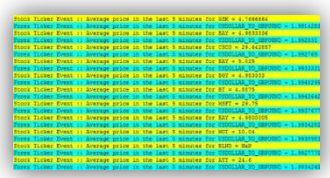
- Follow directly the execution of individual process instances, then abstract from instance to process level:
  - **Active role**: play a specific role, e.g. customer
  - **Passive role**: observe participants and their environment
- Trace business objects in the course of their lifecycle

Potential issues:

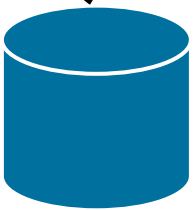
- Active role: no big picture
- Passive role: participants' bias



Automated process discovery and Process mining



event stream

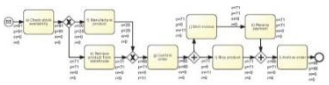


DB



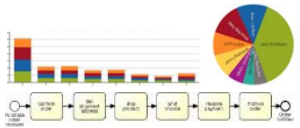
event log

Discovery



discovered model

Enhancement



enhanced model



existing model

Conformance

✔ / ☐

Automated discovery: Minimum data requirements

- Activity name and timestamp
- Reference to case id

case id	event id	properties				
		timestamp	activity	resource	cost	...
1	35654423	30-12-2010:11.02	register request	Pete	50	...
	35654424	31-12-2010:10.06	examine thoroughly	Sue	400	...
	35654425	05-01-2011:15.12	check ticket	Mike	100	...
	35654426	06-01-2011:11.18	decide	Sara	200	...
	35654427	07-01-2011:14.24	reject request	Pete	200	...
2	35654483	30-12-2010:11.32	register request	Mike	50	...
	35654485	30-12-2010:12.12	check ticket	Mike	100	...
	35654487	30-12-2010:14.16	examine casually	Pete	400	...
	35654488	05-01-2011:11.22	decide	Sara	200	...
	35654489	08-01-2011:12.05	pay compensation	Ellen	200	...

Additional information:

- Activity resource, cost
- Case attributes (e.g. customer reference, type of case…)



Exercise 5.3

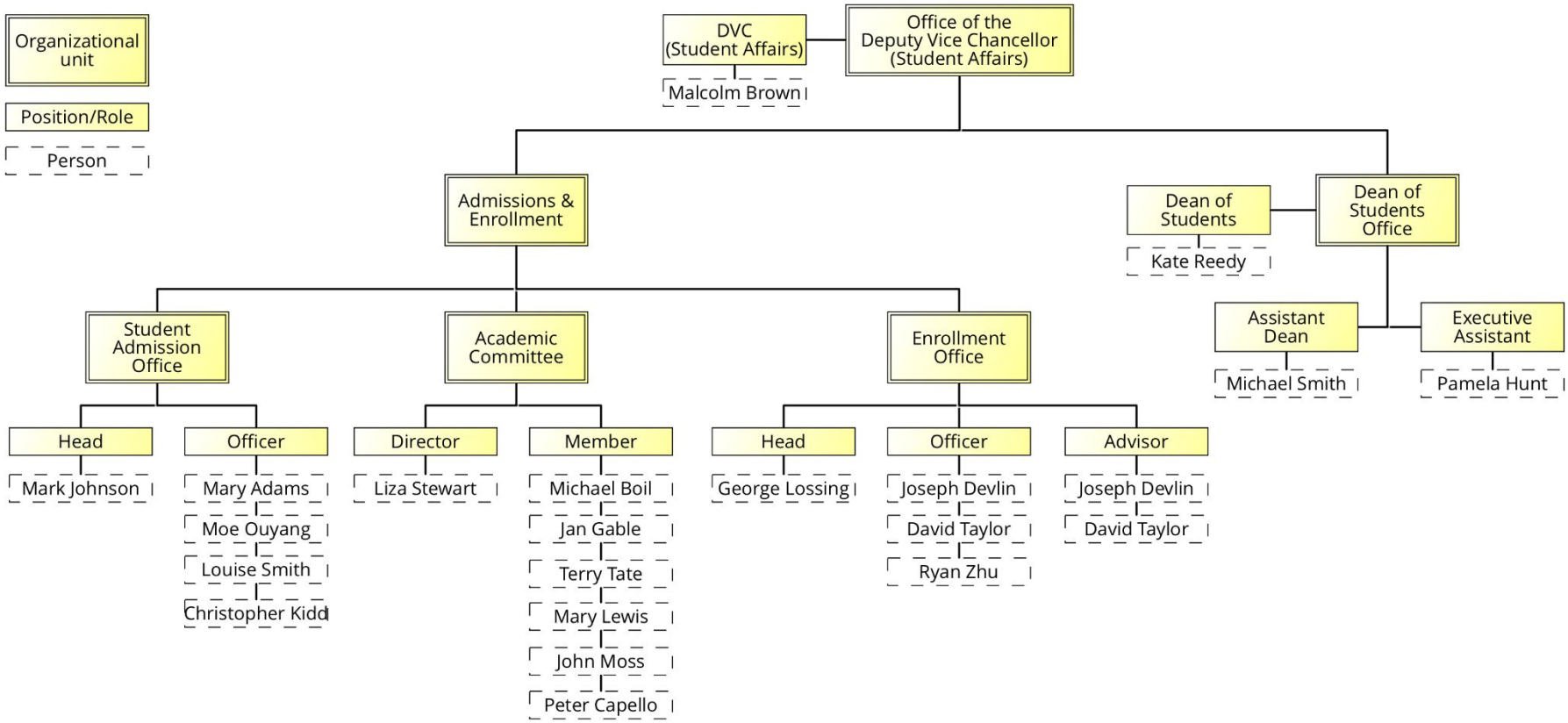
As a process analyst working for the University of Newtown, you have been engaged by Mark Johnson, the process owner of the student admission process, in a project that aims at improving this process. In order to model the as-is process, you start by collecting relevant information about this process. The available documentation includes the organization chart of the Office of the Deputy Vice-Chancellor (DVC) for Student Affairs where Mark's team sits, the UML class diagram of the student admission system which supports this process, and a set of relevant organizational policies that you extract from the university's Web pages.

Based on this documentation, formulate initial hypotheses on how the student admission process works. Next, identify the relevant domain experts to interview and their supervisors whom you should seek approval from.



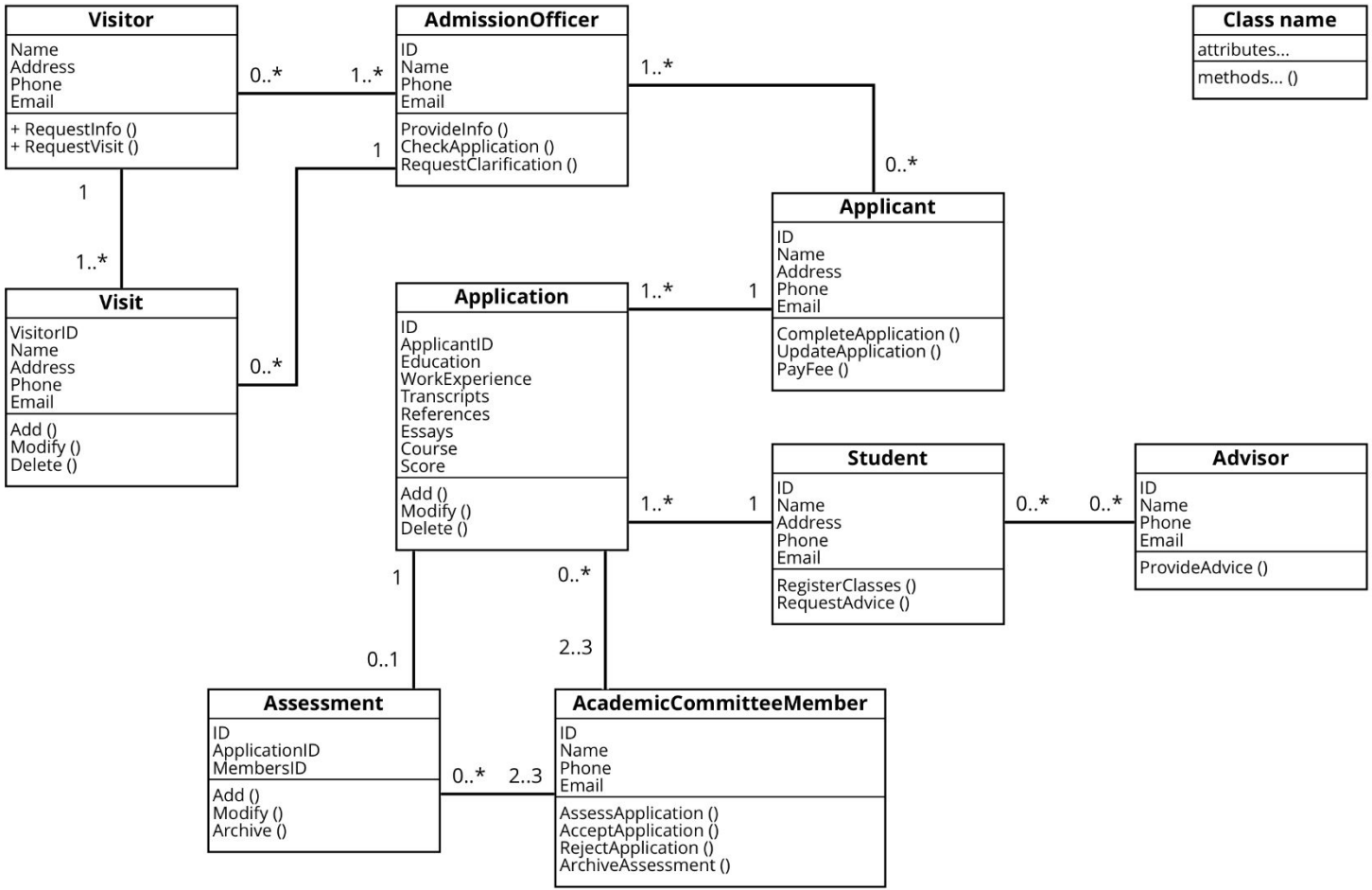
Exercise 5.3 (cont'ed)

Organizational chart of the Office of the DVC (Student Affairs)



Exercise 5.3 (cont'ed)

Extract of the UML class diagram of the student admission system



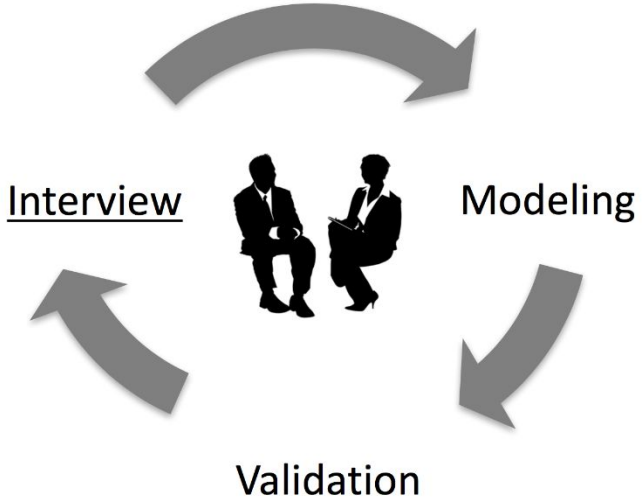


Exercise 5.3 (cont'ed)

Organizational policies for student admission

- *An applicant is admitted if:*
  - *their prior education is consistent with the study area of the selected course*
  - *the submitted essay is not plagiarized and is of good quality*
  - *the score of the prior degree is at least 70 out of 100 (standard 100-point scale)*
  - *the two reference letters are satisfactory.*
- *Successful applicants must accept the offer within four weeks from notification.*

Interview-based Discovery



Approaches:

- forward vs backward
- structured vs unstructured

Assumption: analyst and domain expert share terminology

Pitfall: exceptional behavior tends to be neglected > use questions that aim to identify such behavior



Exercise 5.4

After collecting relevant information on the student admission process, you interviewed some representatives for the two roles involved in this process: Mary Adams and Louise Smith as student admission officers, and Peter Capello as a member of the academic committee (Mark Johnson, the process owner, confirmed that the enrollment office is not involved in this process). The relevant parts of the interview transcripts are provided below.



Exercise 5.4 (cont'ed)

Student admission officer (Mary Adams):

"My process starts when I receive an application for admission. First, I check the completeness of this document. If the application is incomplete, I need to send a request for clarification back to the applicant. Otherwise, I forward it to the academic committee. I then receive a response from the academic committee which can be either of the below:

- A notification of acceptance from the academic committee. In this case, I prepare a letter of offer and send it to the applicant via post to collect his or her signature. Most of the times, I receive a signed offer back from the applicant, but sometimes I don't.
- A notification of rejection. In this case, I send a rejection letter to the applicant via ordinary post.

The problem is that the academic committee is often too slow to reply. I wonder if these academics are just too overloaded with work to care about student admissions..."



Exercise 5.4 (cont'ed)

Student admission officer (Louise Smith):

"When I get a fresh application, it's important that this contains all the required information, including name, address, phone number and email address of the applicant. Unfortunately, the Web portal has many bugs and sometimes lets through incomplete applications, which are a nightmare to rectify! This means going back and forth with the student at least a couple of times. Anyway, once the application is complete, I pass it to a member of the academic committee using our internal student admission system – the same that collects applications via the Web portal. Most of the times, the member of the academic committee replies with a notification of acceptance, in which case I need to prepare a letter of offer and send it to the applicant via post. Our policies are such that applicants must reply within four weeks. In fact, we are flooded by applications, so if they do not hurry to reply, we will offer the place to someone else. Sometimes I receive a notification of rejection. Well, in this case I formulate a rejection letter and send it to the applicant via post."



Exercise 5.4 (cont'ed)

Member of academic committee (Peter Capello):

"When I receive an application from the admission officer, I assess its quality. I extract the grade of the applicant from his or her previous degree and convert it to a standard score based on a conversion table. The score must be at least 70%; otherwise the student is out. Next, I perform a plagiarism check of the essay contained in the application using our pla- giarism detection software. Most of the times, the essay is plagiarism-free. If so, I proceed to read it and assign it a score. Finally, I read the two reference letters attached to the appli- cation. There's a lot you can learn from a reference letter. Often there are subtle messages that the referee wants you to get, like "This is a great student, but I've had better ones". In any case, based on the score, quality of the essay and reference letters, if I deem that the applicant is qualified, I send an acceptance notification to the admission officer, otherwise I send a rejection notification. In either case, I archive the results of my assessment in my database. Ah, I communicate with the student admission office using our internal student admission system. A piece of junk, sometimes messages get lost and I have to send them again, if I'm lucky to find it out!"



Exercise 5.4 (cont'ed)

Next, you took an active role in observing how this process works by acting as the applicant. Using a fake identity (in agreement with the process owner), you triggered this process several times by submitting various applications via the Web portal. Af- ter this, you came up with the following observations.



Exercise 5.4 (cont'ed)

Applicant:

To apply for admission, the applicant needs to prepare an admission application and submit it to the university via a Web portal. The application must include academic transcripts, an essay and two reference letters. The applicant will then receive a response from an admission officer via ordinary mail, which can be:

- A letter of offer. In this case, the applicant needs to sign the letter of offer and return it to the admission officer via post within four weeks.
- A rejection letter. In this case, the applicant does not do anything further and the pro- cess is finished.
- A request for clarification from the admission officer. This is an email notification. In this case, the applicant provides the required documentation to the admission officer by submitting an updated application through the same Web portal used for the initial submission, and then gets a response that is the letter of offer, the rejection letter, or again a request for clarification.



Exercise 5.4 (cont'ed)

Using the information above, create a draft BPMN model of the as-is student admission process. This draft will then be validated with the people that have been interviewed, before sign-off by the process owner. Make appropriate assumptions.

Workshop-based Discovery

- Gather all key stakeholders together
- Participants interact to create shared understanding
- Typically one process analyst (facilitator), multiple domain experts, process owner may also attend
- May be software-supported – a model is directly created during the workshop (typically a separate role – tool operator)
- Model is used as a reference point for discussions
- Alternative: brown-paper workshops
- Usually 3 to 5 half-day sessions



Example: Any difference in discovery?

Consider the following two companies:

- **Company A** is young, founded three years ago, and has grown rapidly to a current toll of one hundred employees.
- **Company B** is owned by the government and operates in a domain with extensive health and security regulations.

How might these different characteristics influence a workshop-based discovery approach?





Discovery and Culture

Before starting with process discovery, it is important to understand the **culture** and the sentiment of an organization.

- There are companies that preach and practice an open culture in which all employees are encouraged to utter their ideas and their criticism. Such organizations can benefit a lot from workshops as participants are likely to present their ideas freely.
- In strictly hierarchical organizations, it is necessary to take special care that every participant gets an equal share of parole in a workshop and that ideas and critique are not held back.

It might be the case that the young dynamic company has a more open culture than the company with extensive health and security regulations. This has to be taken into account when organizing a workshop.



Exercise 5.5

Consider the complaints that have emerged from the interviews reported in Exercise 5.4. As a facilitator, what questions would you ask the various participants to investigate these further in a workshop?

Discovery methods: strengths and weaknesses

Aspect	Evidence-based	Interviews	Workshops
Objectivity	High	Medium-high	Medium-high
Richness	Medium	High	High
Time consumption	Low-medium	Medium	Medium
Immediacy of feedback	Low	High	High

Discovery methods: strengths and weaknesses

Method	Strength	Weakness
Document Analysis	• Structured information • Independent from availability of stakeholders	• Outdated material • Wrong level of abstraction
Observation	• Context-rich insight into process	• Potentially intrusive • Stakeholders likely to behave differently • Only few cases
Automatic Discovery	• Extensive set of cases • Objective data	• Potential issue with data quality and level of abstraction
Interview	• Detailed inquiry into process	• Requires sparse time of process stakeholders • Several iterations required before sign-off
Workshop	• Direct resolution of conflicting views	• Requires availability of several stakeholders at the same time

In what situations is it simply not possible to use one or more of the described discovery methods?



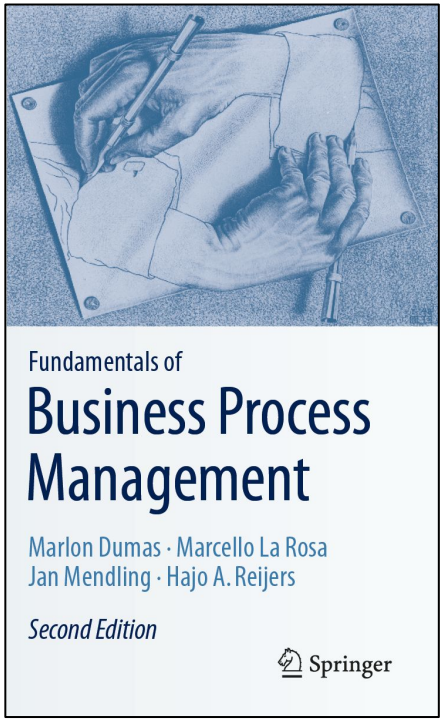


Exercise 5.6

The order-to-cash process of your favorite online bookstore has ten major activities conducted by ten people with five different roles. How much time do you approximately need for creating a process model that is validated by the various stakeholders and approved by the process owner?

Consider two scenarios: one in which you run interviews, the other in which you run workshops. You may also use other discovery methods in these two scenarios, in addition to either inter- views or workshops. Can you estimate the difference in time effort between the two scenarios? Make appropriate assumptions.

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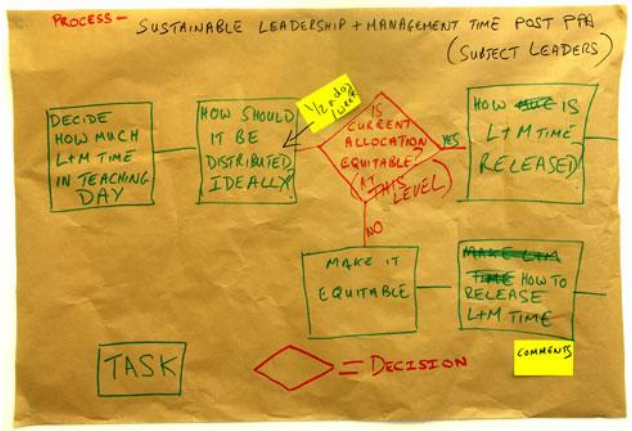


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Stepwise method to conduct the modeling

1. Identify the process boundaries
2. Identify activities and events
3. Identify resources and their handovers
4. Identify the control flow
5. Identify additional elements (e.g. data objects, different types of events, exception handling…)



1. Identify the process boundaries

What are the process triggers?

Purchase order received

What are the possible outcomes (positive/negative)?

Positive outcome: order fulfilled
Negative outcome: order rejected

Which perspective do we assume?

Seller

What artifacts are required as input and output to the process?

Input: Purchase order
Output: Invoice, Shipment notice



Exercise 5.7

Identify the process boundaries for the procure-to-pay process described in Exercise 1.7 of Chapter 1 (page 30).

2. Identify activities and events

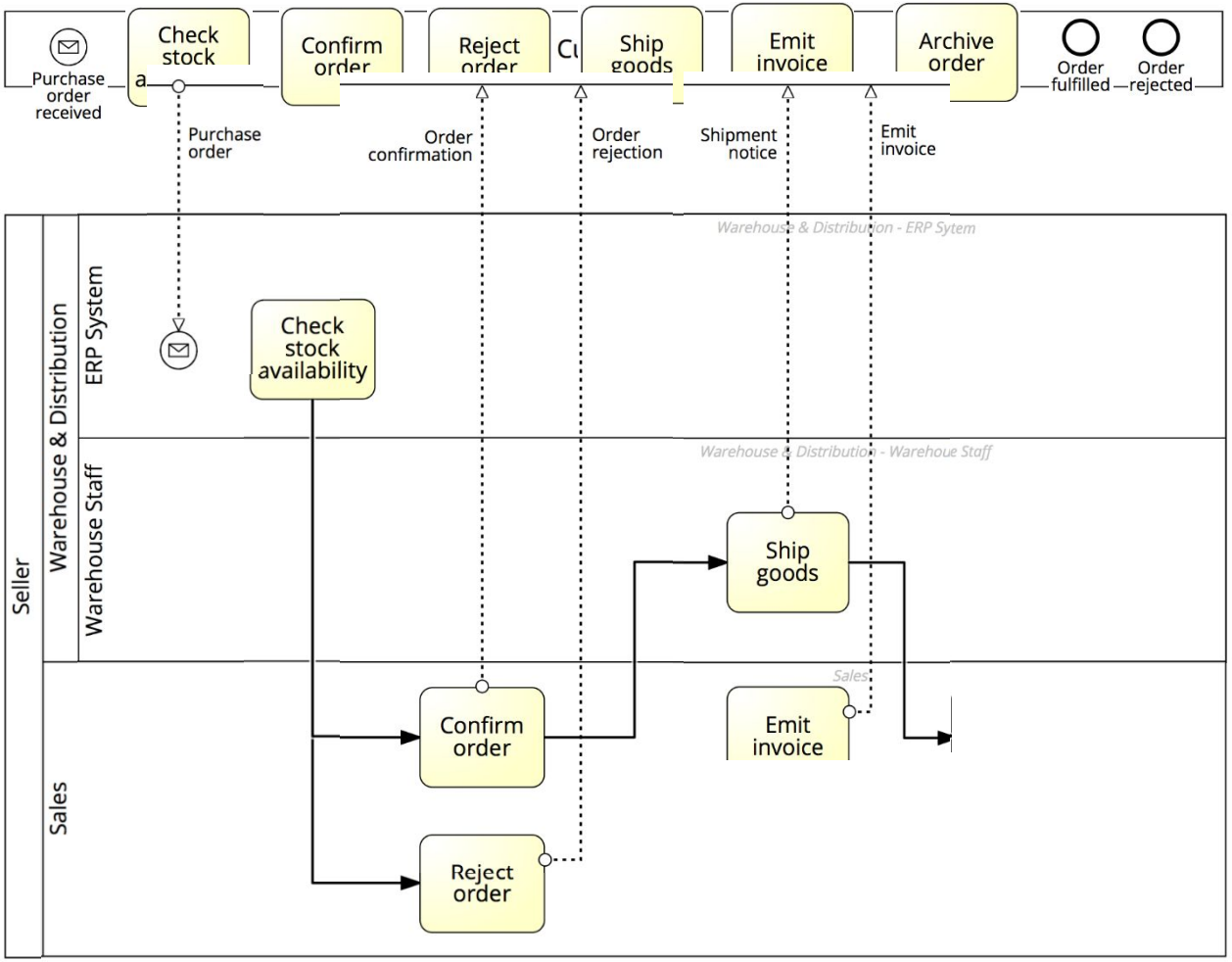




Exercise 5.8

Identify the main activities and events for the procure-to-pay process.

3. Identify resources and their handoffs

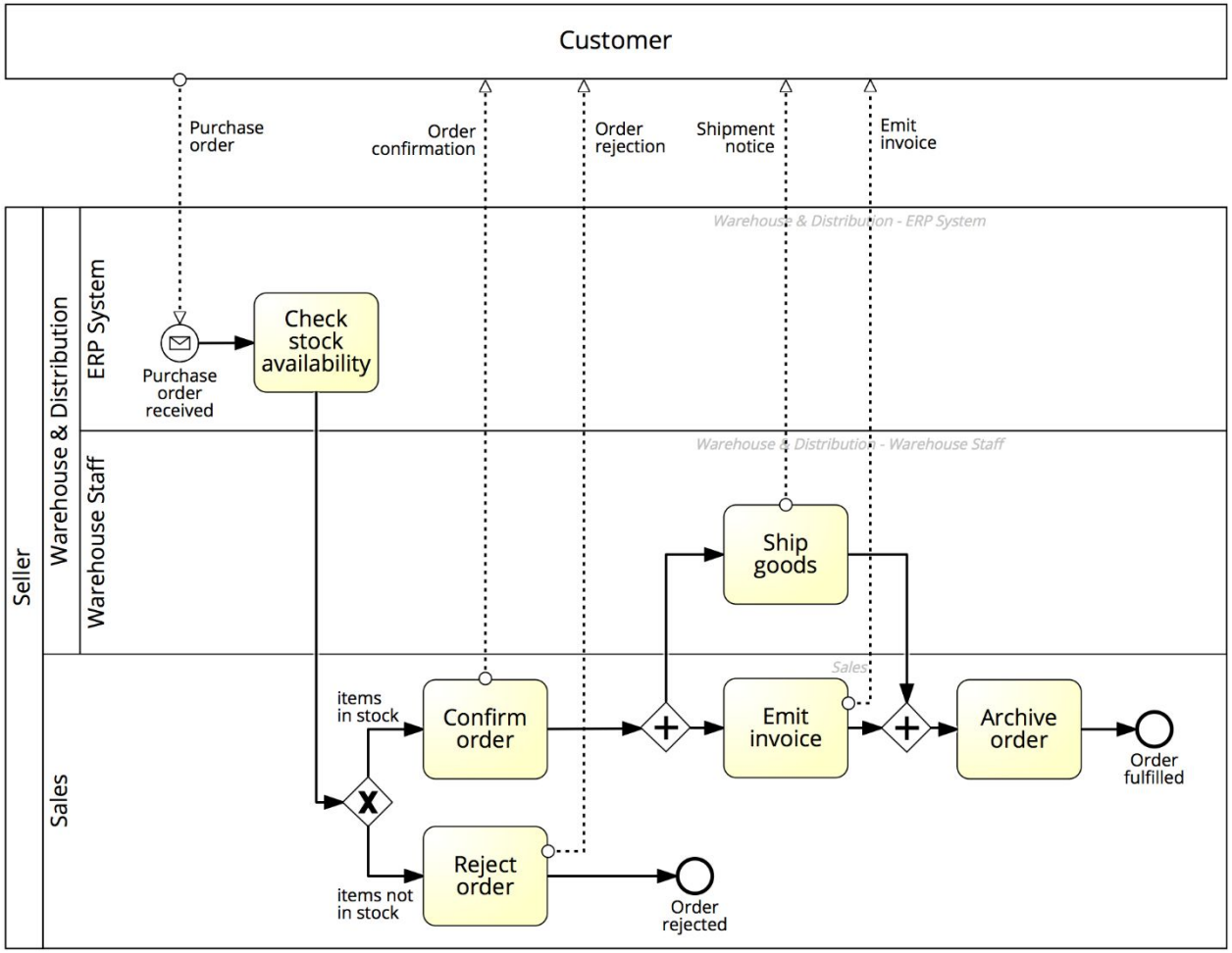




Exercise 5.9

Using the description of the procure-to-pay process, first identify the involved resources; next, assign the activities and events you obtained in Exercise 5.8 to these resources; and finally identify the handoffs.

4. Identify the control flow

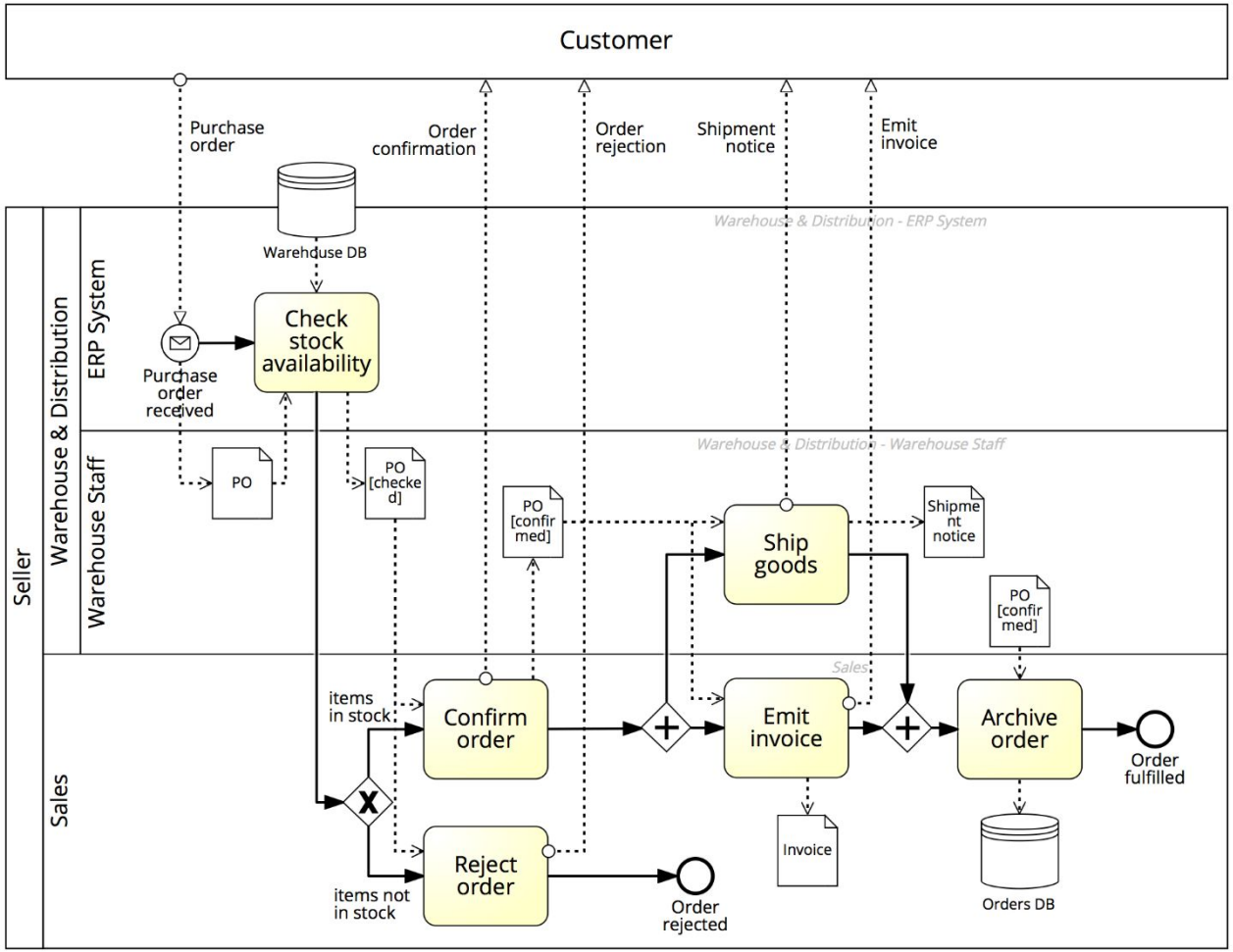




Exercise 5.10

Using the description of the procure-to-pay process, refine the model you obtained in Exercise 5.9 by defining the full control flow.

5. Identify additional elements



When should we stop modelling a process?

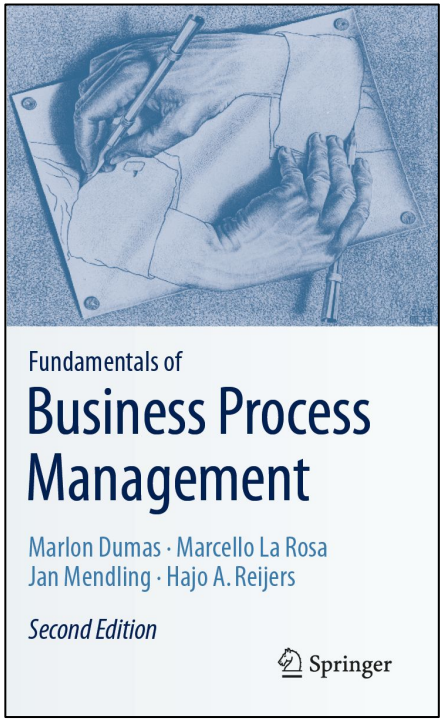




Exercise 5.11

Using the description of the procure-to-pay process, define the model you obtained in Exercise 5.10 by adding business objects and exception handlers.

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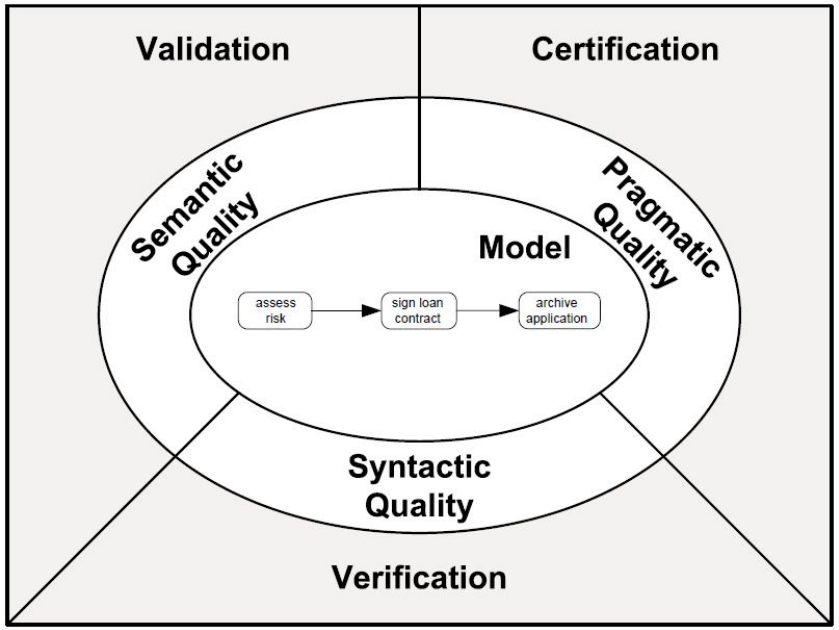


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Process model quality assurance

- Validity
- Completeness



- Understandibility
- Mantainability
- Learning

- Structural correctness
- Behavioral correctness

Syntactic quality: Verification

Syntactic quality relates to the conformance of a process model to the syntactic rules of the modeling language used.

Two types of syntactic rules: structural rules and behavioural rules.

A model is of high syntactic quality if it is syntactically correct:

- Structurally correct +
- Behaviorally correct



Structural correctness

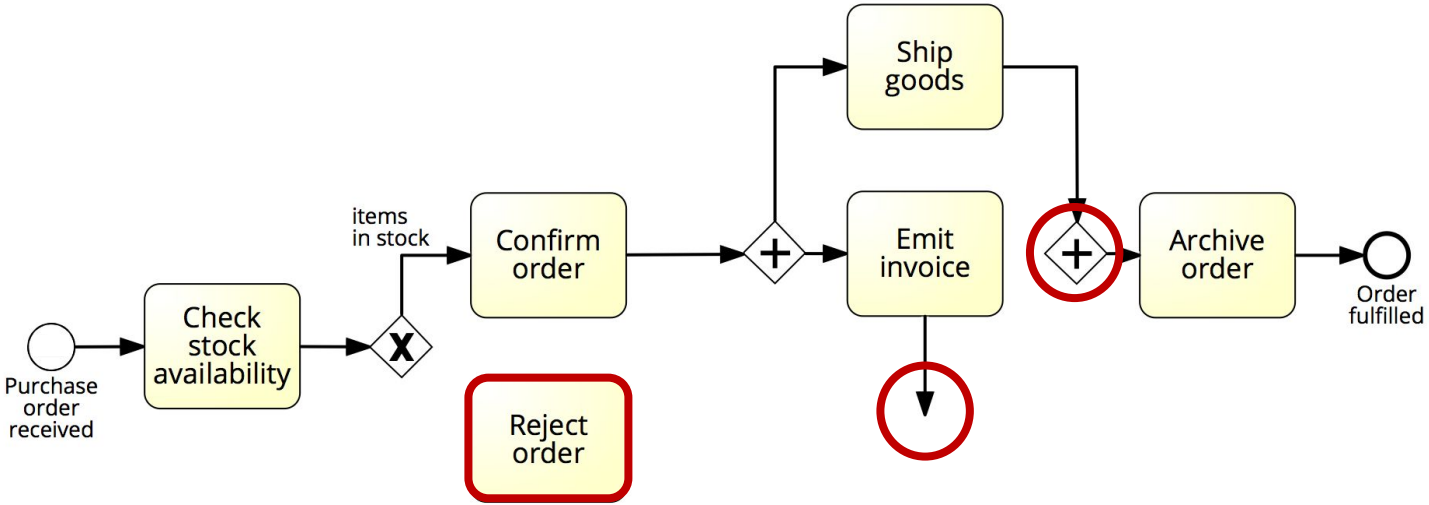
A model is structurally correct if it satisfies the following syntactic rules:

1. **Element-level rules:**
   - activities must have at least one incoming and one outgoing sequence flow
   - start events must not have incoming arcs, end events must not have outgoing arcs
   - gateways must have exactly one incoming and at least two outgoing arcs (splits) or at least two incoming and exactly one outgoing arcs (joins)
   - …

2. **Model-level rule:** all flow nodes must be on a path from a start to an end event
   - i.e. no dangling arcs or disconnected nodes
   - implies that a model should have at least one start and one end event.



Example: structural correctness





Behavioral correctness (aka “soundness”)

A model is sound if it satisfies the following behavioral rules:

1. *option to complete*: any running process instance must eventually complete,

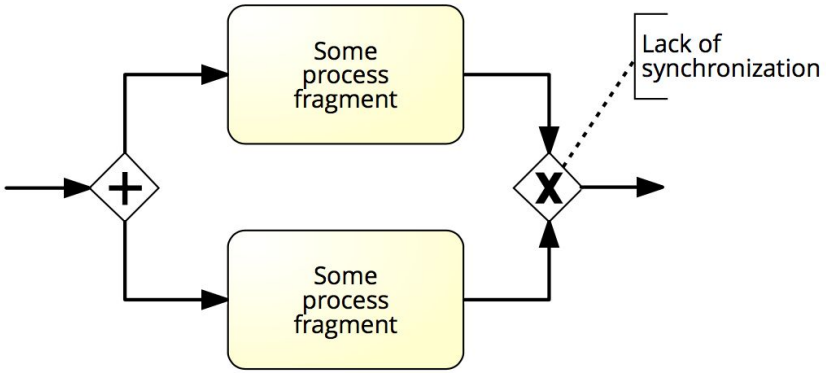
2. *proper completion*: at the moment of completion, each token of the process instance should be in a different end event, and

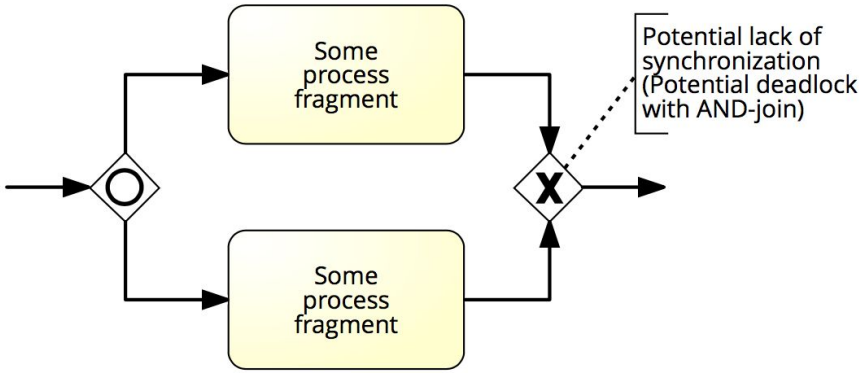
3. *no dead activities*: any activity can be executed in at least one process instance.

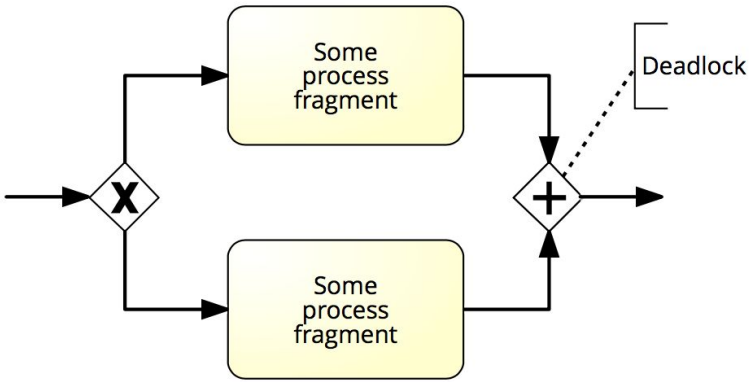


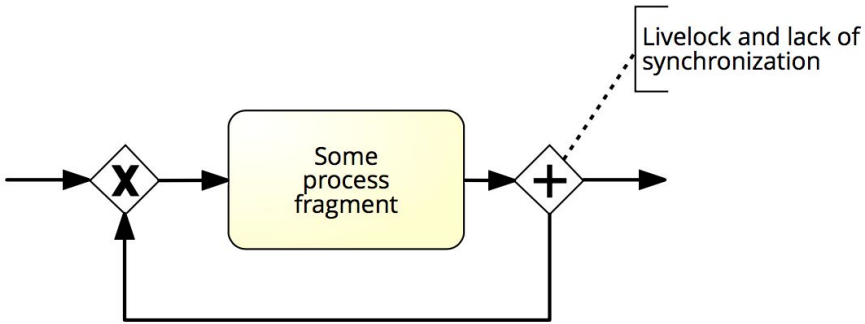
Exercise 5.12

Have a look at Figure 5.11. What is precisely going wrong in each block structure?



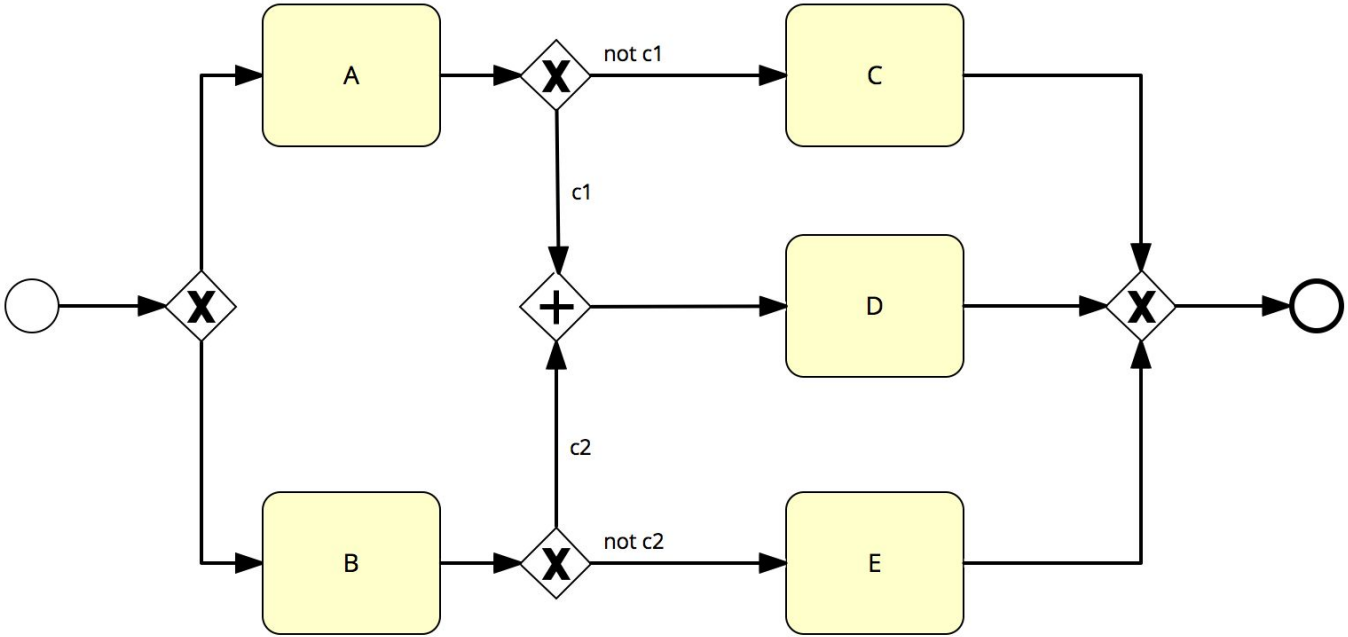






Example: no option to complete

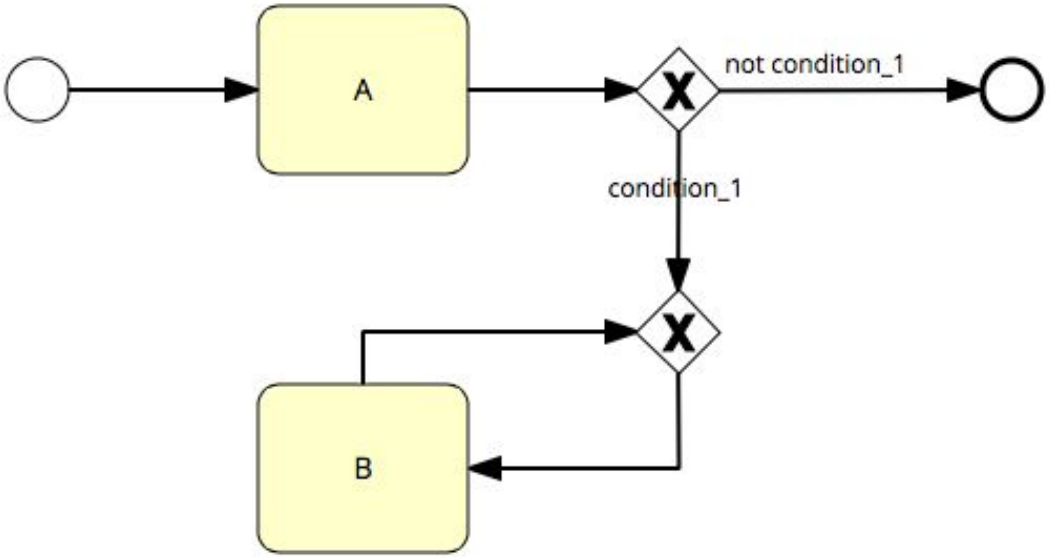
If c1 is true after executing A, or c2 is true after executing B, the instance cannot complete (deadlock)



Note: this model also suffers from a dead activity (D)

Example: livelock (no option to complete)

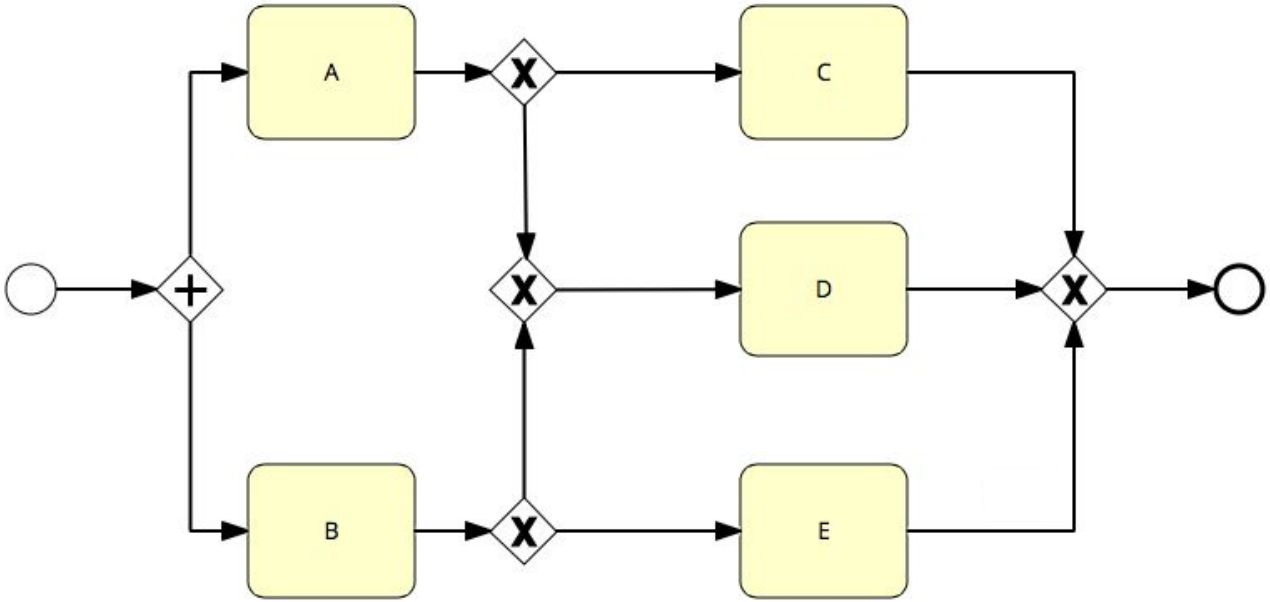
If condition_1 is true, the instance cannot complete and activity B will be repeated forever (livelock)



Note: this model is **structurally incorrect**, because B is not on a path to the end event

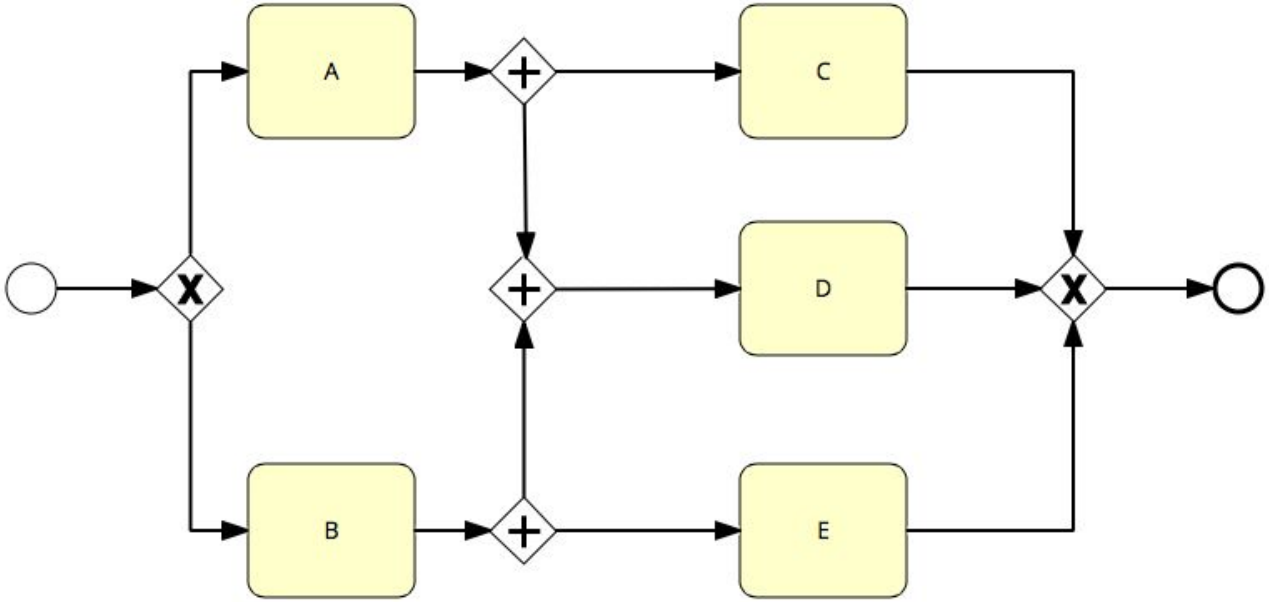
Example: no proper completion

At the moment of completion, there will be two tokens in the end event (lack of synchronization)



Example: dead activity

Even if this model can always complete, Activity D will never be executed

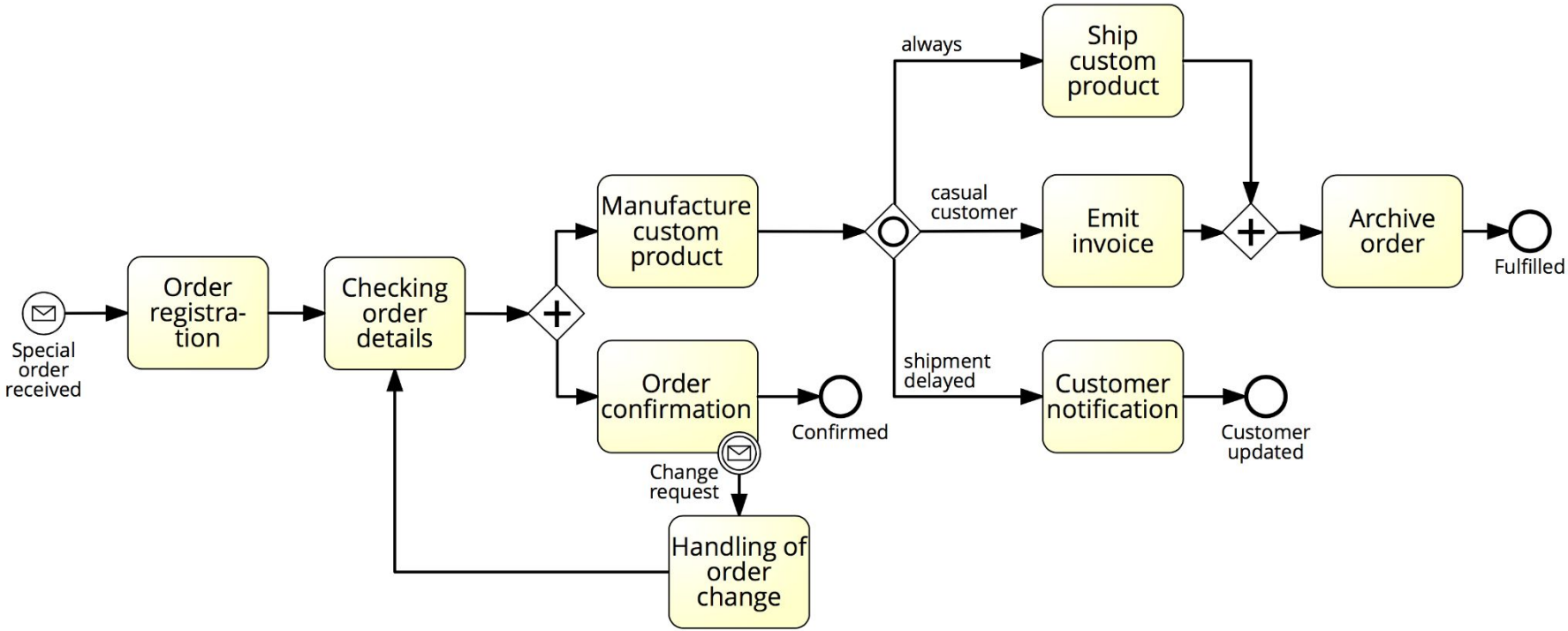


Note: this model also suffers from a deadlock, as a token will be left behind (stuck before the AND-join). However, there is always an option to complete

Exercise 5.13

Fulfilment of special orders

- Which behavioral rules are violated in the model below?
- How can this model be made sound?



Semantic quality: validation

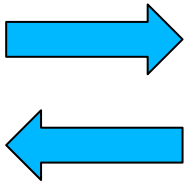
Semantic quality relates to the adherence of a process model to its real-world process.

Validation is the activity of checking the semantic quality of a model by comparing it with its real-world business process.

A model is of high semantic quality if it is semantically correct:

- Valid (all *model* instances are correct and relevant) +
- Complete (all possible *process* instances are covered)







Domain Expert

Process Analyst



Semantics

Meaning of the various elements

- Activities model something actively performed during the process
- Events model something happening instantaneously during the process
- AND gateways model parallelism
- XOR gateways model exclusive decisions and simple merging points
- OR gateways model inclusive decisions and synchronizing merging points

+

Meaning of the whole business process model

- *"This model captures an order fulfilment process that takes place at a seller. The model starts with the receipt of an order…"*



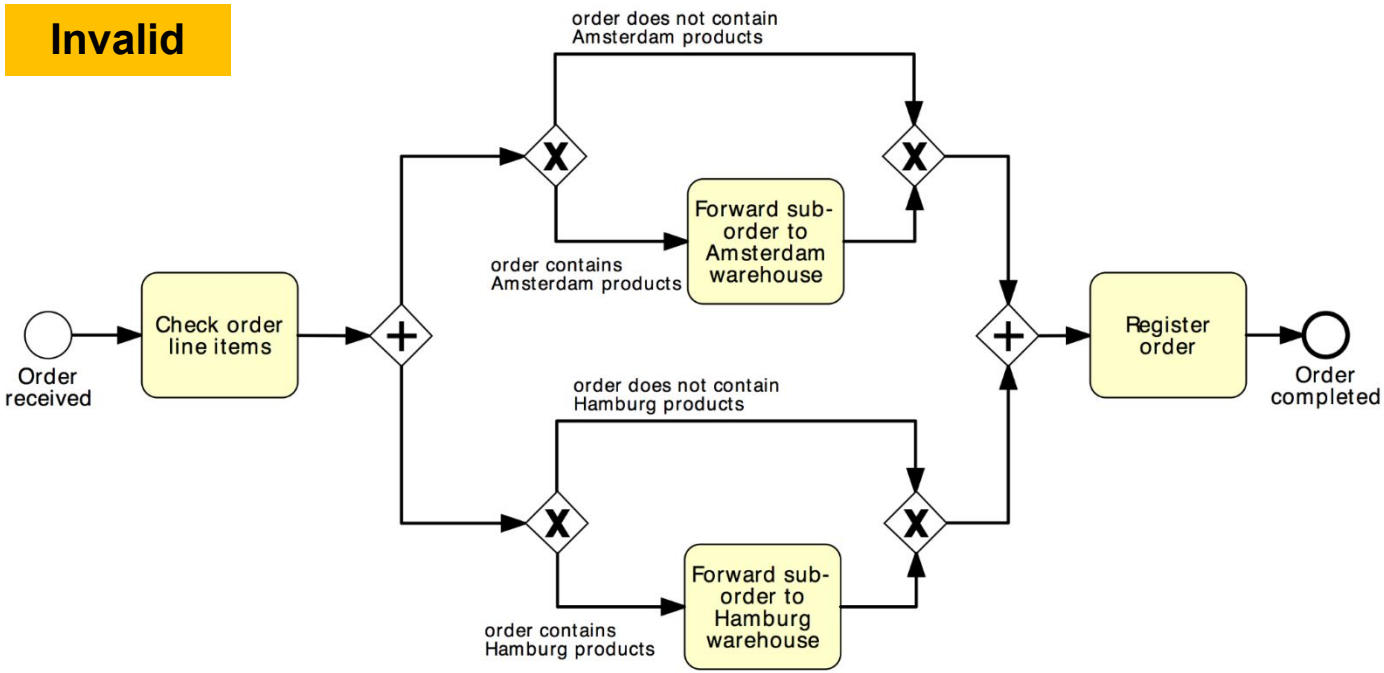
Exercise 5.14

Consider the following process description.

A company has two warehouses that store different products: Amsterdam and Hamburg. When an order is received, it is distributed across these warehouses: if some of the relevant products are maintained in Amsterdam, a sub-order is sent there; likewise, if some relevant products are maintained in Hamburg, a sub-order is sent there. Afterwards, the order is registered and the process completes.

Exercise 5.14 (cont'ed)

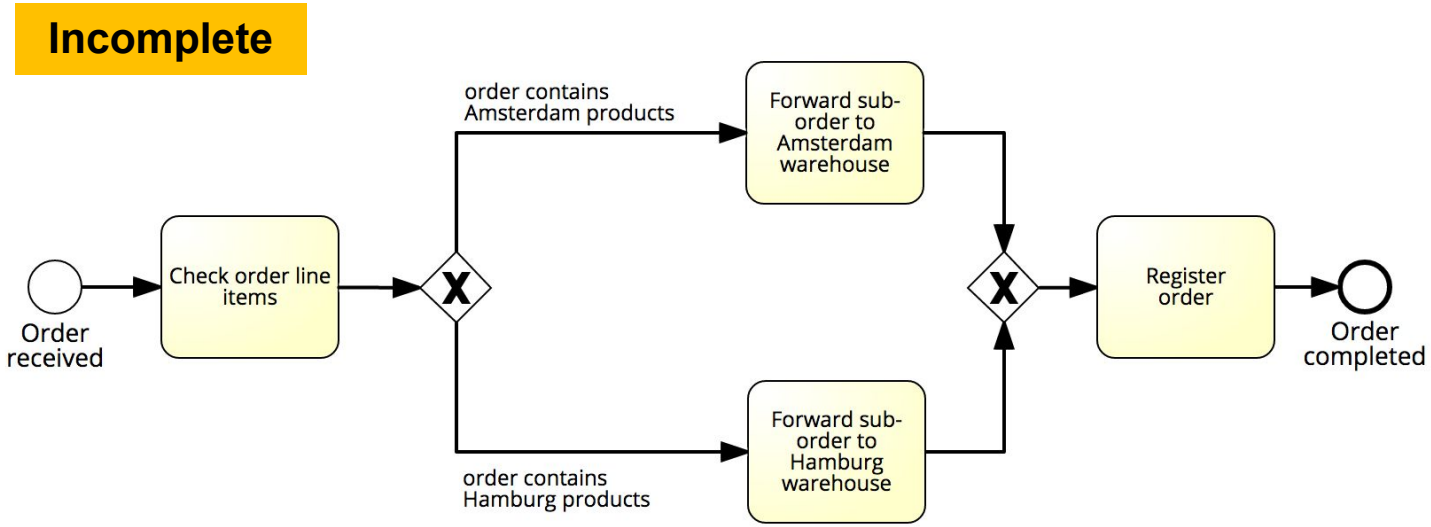
What can we say about the semantic quality of this model?



It is **not possible** that products are neither in the Amsterdam nor in the Hamburg warehouse.

Example: semantic correctness

And about this one? (with reference to the same description)



Orders may contain **both** products that are in Amsterdam and products that are in Hamburg



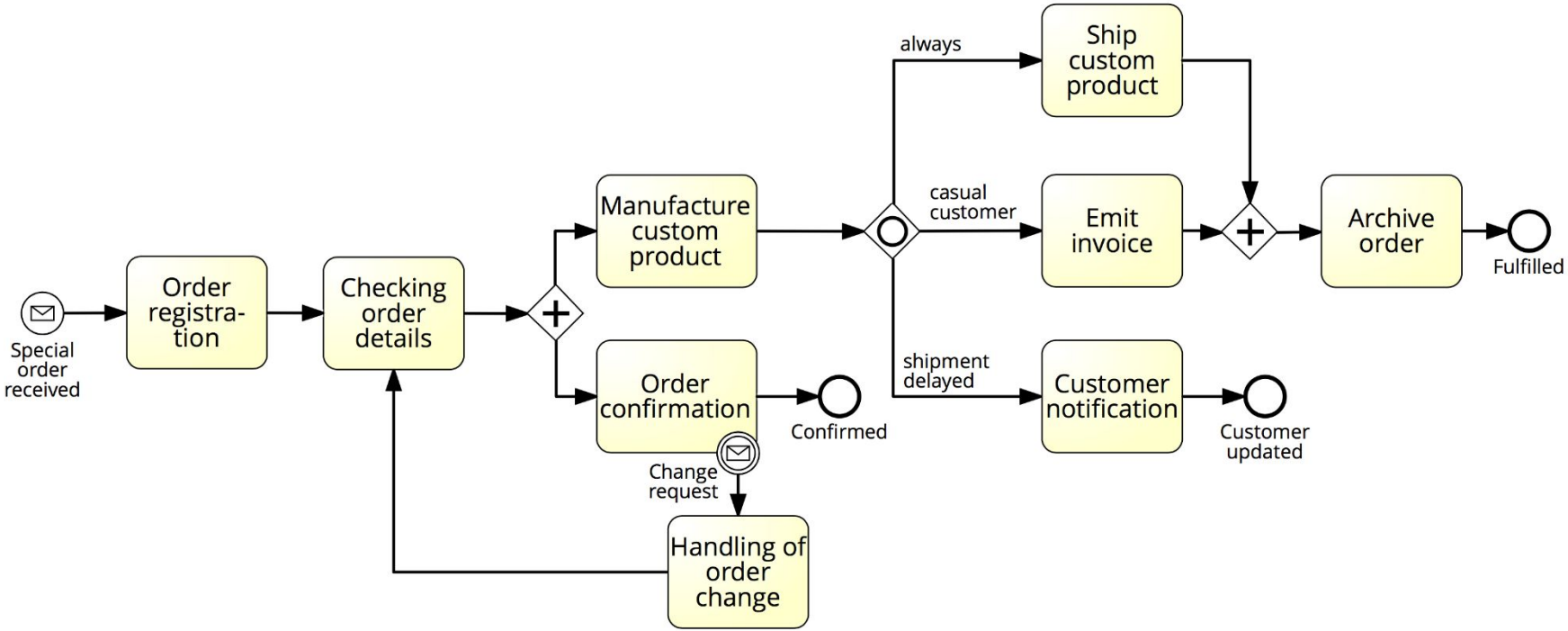
Exercise 5.15

Consider the model in the next slide, with reference to the following process description. Is this model valid and complete? If not, what statements are invalid and what is missing?

When a special order is received, it is first registered and then its details are checked. Next, the order is confirmed and meantime the custom product is manufactured. Once the product has been made, the shipment can be planned. Afterwards, the customer type and shipment status are checked. In fact, if a customer is casual an ad hoc invoice must be emitted, which is not required for ordinary customers. In the latter case, the customer account is simply charged with the costs related to the order fulfillment. Moreover, if the shipment has been delayed, the customer must be updated on the expected delay. Concomitantly to these activities, the custom product is shipped. After the latter activity and after the invoice has been emitted, the process completes with the archival of the order. Any time during the confirmation of the order and the manufacturing of the respective product, an order change request may be received, in which case any activity must be interrupted to handle the change re- quest. This includes the registration of the order variation and a notification to the customer, after which the process resumes from the order checking.



Exercise 5.15 (cont'ed)





Pragmatic Quality: Certification

Pragmatic quality relates to the usability of a process model

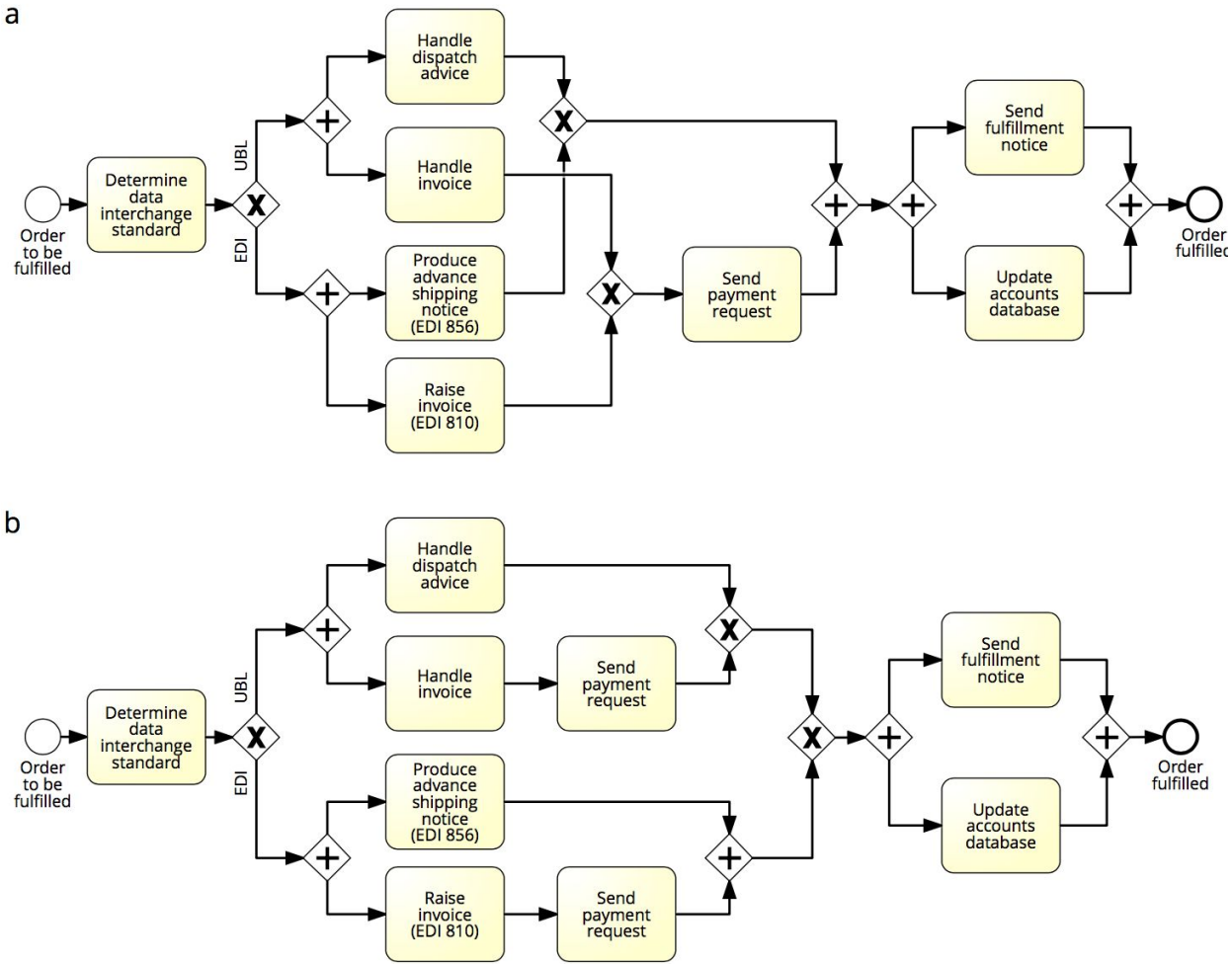
Challenge = anticipate the particular usage of the model

Usability:

- Understandability: how easy it is to read and comprehend the model
- Maintainability: how easy it is to apply changes
- Learning: how good a model reveals how its corresponding process works in reality

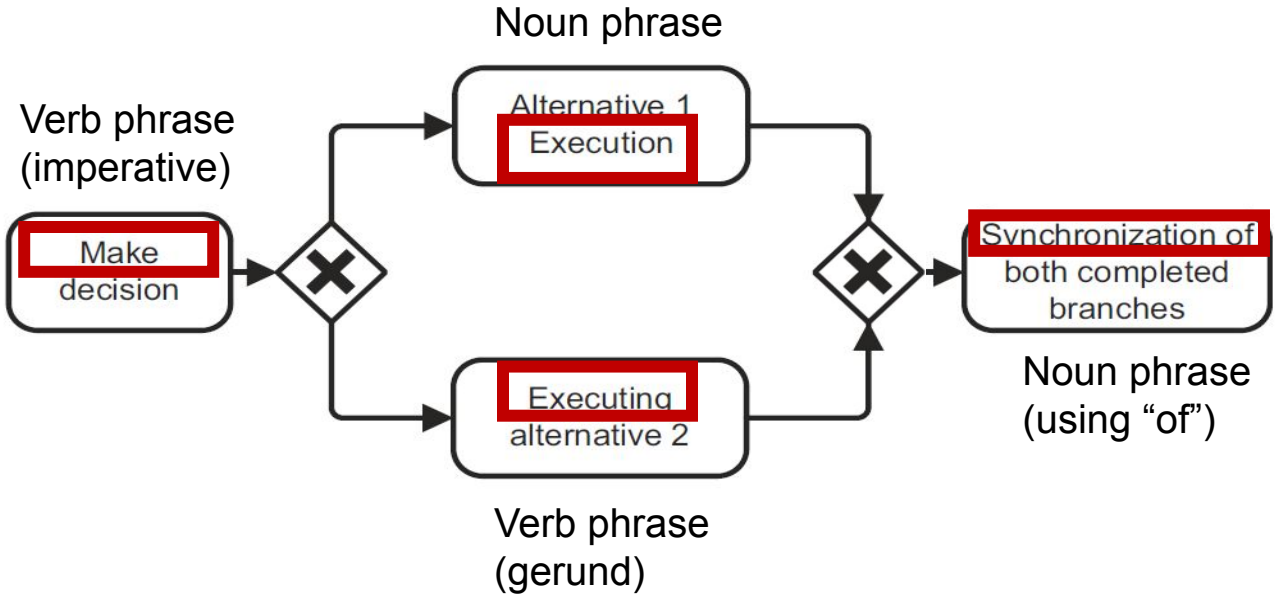
Model characteristics that influence usability include size, structural complexity and layout

Example: block-structuring a model

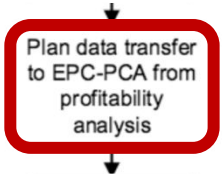


Is this process model of good pragmatic quality?

Different labeling styles



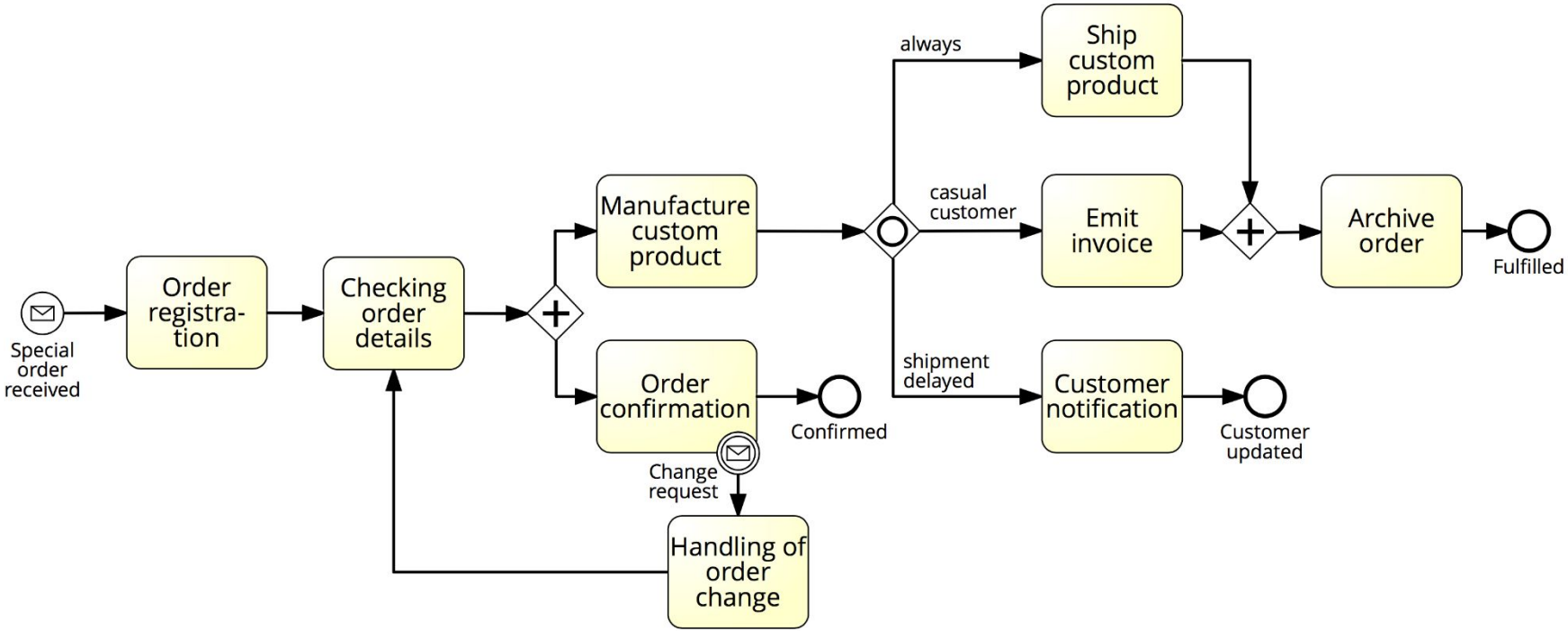
Example: A model from the SAP R/3 collection…





Exercise 5.16

Is the process model below of good pragmatic quality? If not, how can it be improved?



Modeling Guidelines and Conventions

Used to improve **pragmatic quality** by restricting:

1. Vocabulary (e.g. banning the use of the OR gateway)
2. Structure (e.g. limiting the size of a model to max 30 nodes)
3. Semantics – rare (e.g. using data objects to only capture information flow)
4. Appearence (e.g. enforcing a particular labeling or layout style)

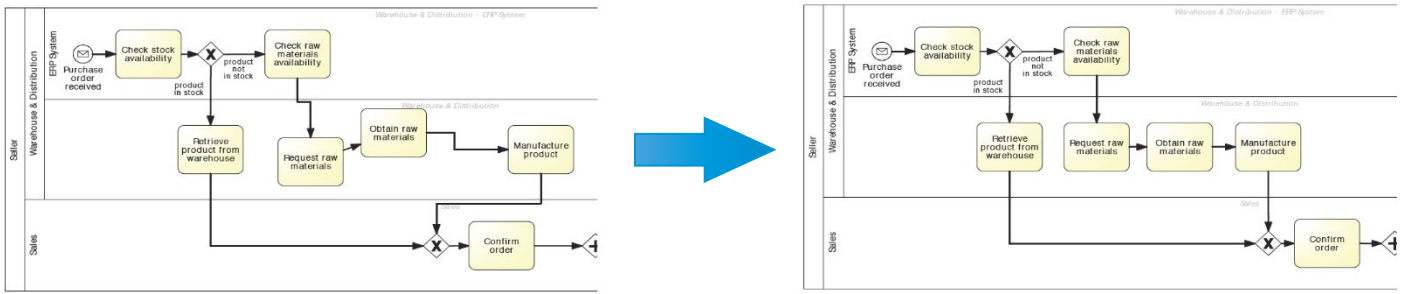
Example: modeling guidelines and conventions

Labeling

1. Activities as imperative verb + noun
2. Events as noun + past-participle verb
3. Conditions on outgoing arcs of (X)OR-splits with reference to object

Layout

1. From top-left to bottom-right
2. Use direct arcs with no crossing where possible



Example modelling guidelines: 7PMG

G1 Use as few elements in the model as possible

G2 Minimize the routing paths per element

G3 Use one start event (per trigger) and one end event (per outcome)

G4 Model as structured as possible

G5 Avoid OR gateways where possible

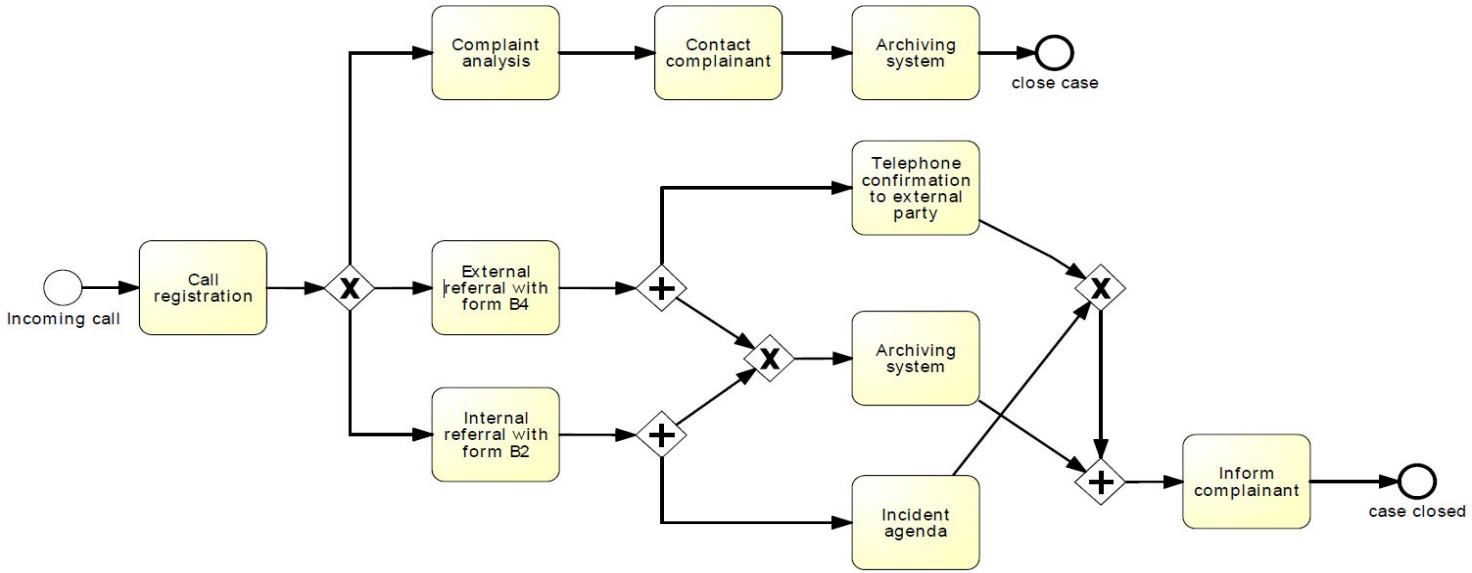
G6 Use verb-object activity labels

G7 Decompose a model with more than 30 elements

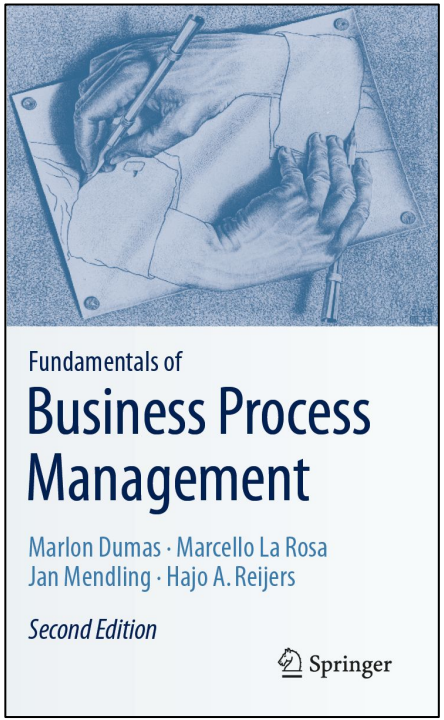
Exercise 5.17

Which 7PMG can be applied to this model? Consider the description below.

A complaint is triggered by a phone call by a complaining customer. It is decided whether the complaint can be handled or whether it has to be referred to an internal or external party. An external referral leads to a telephone confirmation to the external party. An internal referral is added to the incident agenda. If no referral is needed, a complaint analysis is conducted and the complainant is contacted. In either case, the complaint is archived and the case is closed.



Chapter 5: Process Discovery



Contents

1. The Setting of Process Discovery
2. Process Discovery Methods
3. Process Modeling Method
4. Process Model Quality Assurance
5. Recap

Recap

1. For Process Discovery, we have to define the setting, gather information, conduct modeling and assure model quality
2. There are various discovery methods such as document analysis, observation, automated process discovery, interviews and workshops
3. A modeling method proceeds by identifying the process boundaries, then activities and events, resources and their handoffs, control flow and additional elements
4. Quality assurance relates to syntactic, semantic and pragmatic quality

